

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Tutorial 3

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- Consider the electrical circuit shown in Figure 1. Evaluate the voltage v_x across the resistor R_3 using nodal analysis. Use the following numerical values of the circuit elements

$$V_1 = 100V, V_2 = 150V, I_1 = 5A, I_2 = 10A$$

$$R_1 = 20\Omega, R_2 = 10\Omega, R_3 = 25\Omega, R_4 = 12.5\Omega$$

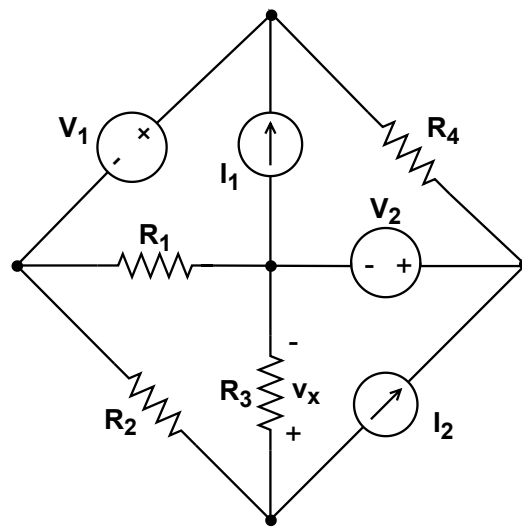


Figure 1: Electrical circuit with independent voltage and current source. Evaluate v_x .

- Consider the electrical circuit shown in Figure 2. Evaluate the mesh currents i_1, i_2, i_3 and the voltages V_{I1}, V_{I2} across the respective current sources I_1 and I_2 . Use the following numerical values of the circuit elements.

$$V_1 = 4V, V_2 = 5V, V_3 = -11V$$

$$I_1 = k_1 \times i_x = 2i_x, I_2 = k_2 \times v_x = \frac{v_x}{12}$$

$$R_1 = 2\Omega, R_2 = 3\Omega, R_3 = 1\Omega, R_4 = 1\Omega, R_5 = 4\Omega$$

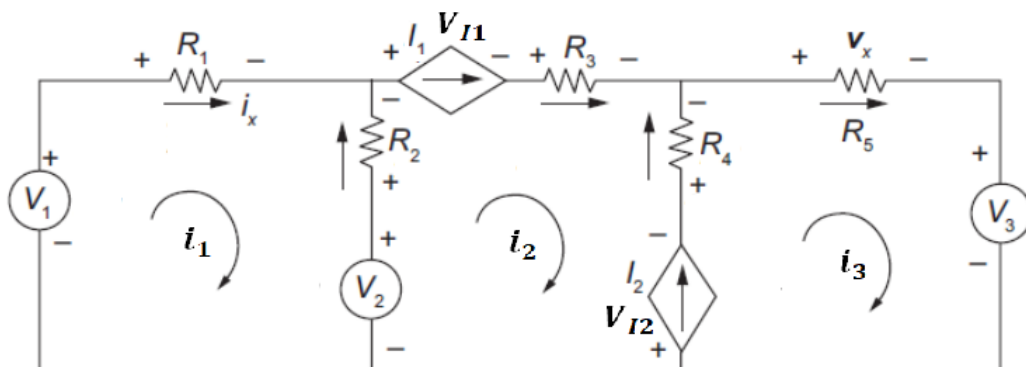


Figure 2: Electrical circuit with dependent and independent sources.

3. The following questions relate to base 2, base 3, base 8, base 10 and base 16 representation of numbers. Note that the alphabet for base 16 is 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.

(a) Compute the base conversions as specified in the table:

Base	Number	New Base	Number
10	65/7	2	
10	72/7	3	
8	4567	16	
16	10AF	8	

- (b) Find the 4 most significant bits in the base 2 representation of $(\sqrt{2})_{10}$.
4. The following questions are related to the base 2 and base 3 representation of numbers.

(a) A magician shows the audience the following cards:

Card 1	1	3	5	7	9	11	13	15
Card 2	2	3	6	7	10	11	14	15
Card 3	4	5	6	7	12	13	14	15
Card 4	8	9	10	11	12	13	14	15

The audience secretly picks a number between 1 and 15 inclusive, and then hands the magician the cards on which this number appears. The magician is then able to tell the audience right away what that number is. For example, if the audience's number is on cards 1, 3 and 4 but not card 2, the magician can compute the number by adding the first numbers on cards 1, 3 and 4, i.e. $1 + 4 + 8 = 13$. Explain how this trick works. (Hint: base 2 representation).

- (b) Among 9 coins, there are 8 real ones and a fake one. All real coins have the same weight, which is greater than the weight of the fake coin. In two weighing on a balance, identify the fake coin. (Hint: base 3 representation).